

MD ABDUS SHAKUR SIAM

shakursiam@gmail.com | Phone: (+880) 1916900901 | [LinkedIn](#) | [Website](#)

RESEARCH INTERESTS

Embedded Systems, Cyber-Physical Systems, AIoT and Edge Intelligence, IoT Network Security and Privacy, Secure Sensing Platforms, Battery Management Systems

EDUCATION

BSc | *Electrical and Electronics Engineering* CGPA: 3.28/4.00
Ahsanullah University of Science and Technology

Jun 2019 – November 2023
Dhaka, Bangladesh

Thesis: Analysis of the flight control system for designing a subsonic sounding rocket for meteorological particle observation.

Relevant Coursework: VLSI Design, Computer Architecture, Control Systems, Multimedia Communication, Processing and Fabrication Technology

RESEARCH EXPERIENCE

Research Assistant (*Embedded Systems & AIoT*)

Department of EEE, Bangladesh University of Engineering and Technology

August 2024 – Present

HIVE (Hydrogen, Intelligent Vehicle & Energy) Research Group

Supervisor: Dr. Muhammad Anisuzzaman Talukder

- Designed and fabricated the "Edge-SBC" custom hardware platform integrating a BQ24610 buck charger with an ESP32-S3 host for real-time voltage, current, and thermal telemetry.
- Deployed a Float32-downcasted ML model for Battery SoH/RUL estimation, achieving 52 μ s steady-state latency, ≤ 21 μ J energy cost, and 1.80% MAE in live Physical-in-the-Loop (PhIL) validation.
- Conducted a 32-experiment cross-dataset generalization protocol across 4 public benchmarks and 3 battery chemistries (NCA, LFP, LCO), establishing a protocol-affinity transfer ceiling of $R^2 = 0.761$.
- Led a co-authored survey synthesizing 168 AI-IoT security studies into a 4D taxonomy, identifying critical gaps in cross-domain generalization and formal privacy guarantees.

Cooling Paint vs. Conventional Paint Performance Evaluation (*Industry-Funded: Berger Paints Bangladesh Ltd.*)

- Developed an ESP32/DHT22 IoT sensing network across 3 pilot enclosures for continuous thermal monitoring in tropical monsoon conditions (Köppen Am).
- Demonstrated a peak midday internal temperature reduction of 4.15°C (10.74%) and ~5°C Heat Index suppression, corroborated by 8.0°C surface reduction in lab step-heating tests.
- Led experimental data analytics, CIELAB colorimetric degradation modeling, and journal manuscript drafting.

Research Associate (*Consultancy*)

Infrastructure Investment Facilitation Company (IIFC)

April 2026 – Present

- Contributing to national scientific inventory assessment report for Ministry of Finance

MANUSCRIPTS IN PREPARATION & PREPRINTS

- Md. Abdus Shakur Siam and Muhammad Anisuzzaman Talukder. "Edge-Deployed Battery State-of-Health Estimation on a Custom Synchronous Buck Charging Platform: A Cross-Dataset Lightweight Machine Learning Framework." [Under Final Review | Target Journal: Elsevier Journal of Energy Storage]
- Md. Abdus Shakur Siam, Shamik Dey Deepto, Sabit Shahriar Haque, Omar Farrok. "Security, Privacy, and Secure Networking in AI-based Internet of Things Systems: A Comprehensive Survey." [Under Review at: Journal of Network and Computer Applications]

- Md. Abdus Shakur Siam, Md. Fahim Hossain, Ahmed Imtiaz, A. S. M. Obaidullah Mahmud, and Muhammad Anisuzzaman Talukder. "Multi-Seasonal Assessment of Nanostructured Cool Paints for Passive Radiative Cooling in Tropical Climates Using IoT Sensor Networks." [Manuscript Complete; Pending Submission | Target Journal: Elsevier Energy and Buildings]
- Sayed MD Tashfi Nowroz, Nermeen Saleh, Abdus Shakur Siam, Fathi Amsaad. "A Benchmark Reference for ESP32-CAM Module." Preprint available at ResearchGate: <https://doi.org/10.13140/RG.2.2.33753.02409>

PROFESSIONAL EXPERIENCE

Electrical Hardware Engineer (UK Based-Remote, Part-time)

LABTRONICS DESIGN LIMITED

May 2025 – May 2026

- Designed production-grade multi-layer PCBs (DipTrace) with DFM/DFT compliance for manufacturing.

Embedded Engineer (Germany Based-Remote, Part-time)

ERAVEND GMBH & CO. KG

October 2024 – April 2025

- Designed KiCad-based multi-layer PCBs for a Choco 3D-printer onboard computer; contributed to VTOL drone power electronics.

Executive, Research Development and Production Engineer (Full-time)

FRONTECH LIMITED

October 2023 – July 2024

- Led full-cycle development of the JRC AgriTech Board v3.0, including 4-layer PCB design and sensor integration.

SELECTED PROJECTS

- Ultra-Compact G-Force Alerting and Qi-Enabled Embedded System
- Hybrid Tracking and Remote-Control Embedded Hardware Platform
- Portable Wireless Charging Transmitter (bq500212a-Based)
- MCU-Controlled Smart Battery Charging System (BQ24610 + ESP32-S3)
- Custom ESP32-S3 Embedded Controller for Secure IoT Applications
- Wearable Body Camera Embedded System using STM32H743 and CMOS Image Sensors

TECHNICAL SKILLS

Programming: C/C++, Python, MATLAB

ML / Embedded AI: TensorFlow, TensorFlow Lite Micro, scikit-learn (SoH/RUL modeling, edge deployment)

Embedded Platforms: STM32, ESP32/ESP-IDF, Arduino, Raspberry Pi

PCB & Circuit Design: KiCad, DipTrace, LTspice, OrCAD PSpice, Proteus

Communication Protocols: SPI, I²C, UART, CAN, MQTT, NFC, RFID

IoT & Cloud: AWS IoT Core, Node-RED, LoRaWAN, Wi-Fi, Bluetooth

Version Control: Git

TEACHING & ACADEMIC SERVICE

- Technical Trainer (PCB Design), NAS Tech Engineering; Workshop Instructor, AUST.
- Delivered IoT/embedded design workshops under the EMK Small Grant and at Faridpur Engineering College.

LEADERSHIP & ACADEMIC ACTIVITIES

Vice President, AUST Innovation and Design Club (2023)

- Advanced from Web Developer to VP; managed 90+ volunteers and directed major technical initiatives, VLSI career seminars, and the flagship festival MindSparks'23.
- Raised BDT 1.2M (1200K) in funding through strategic marketing, corporate outreach, and event planning.

Chief Coordinator, TechnoXian Bangladesh (2023 – 2024) - Directed a 50+ member organizing team.